

**Problem**

Find all positive integers  $d$  such that  $d$  divides both  $n^2 + 1$  and  $(n + 1)^2 + 1$  for some integer  $n$ .

*Source: Original Problem by Aryan Goutam*

**§1 Extracting the Hidden Common Divisor**

Suppose that a positive integer  $d$  satisfies

$$d \mid (n^2 + 1)$$

and

$$d \mid ((n + 1)^2 + 1)$$

for some integer  $n$ .

Expanding the second expression gives

$$(n + 1)^2 + 1 = n^2 + 2n + 2,$$

so we have

$$d \mid (n^2 + 1) \quad \text{and} \quad d \mid (n^2 + 2n + 2).$$

A natural step is to subtract the two divisible quantities:

$$d \mid ((n^2 + 2n + 2) - (n^2 + 1)),$$

hence

$$d \mid (2n + 1).$$

Since  $d$  divides  $2n + 1$ , it also divides

$$(2n + 1)^2 = 4n^2 + 4n + 1.$$

On the other hand, because  $d \mid (n^2 + 2n + 2)$ , multiplying by 4 yields

$$d \mid 4(n^2 + 2n + 2) = 4n^2 + 8n + 8.$$

Subtracting again,

$$d \mid ((4n^2 + 8n + 8) - (4n^2 + 4n + 1)),$$

so

$$d \mid (4n + 7).$$

Now we combine the two linear divisibility relations:

$$d \mid (4n + 7) \quad \text{and} \quad d \mid 2(2n + 1) = 4n + 2.$$

Therefore,

$$d \mid ((4n + 7) - (4n + 2)),$$

which gives

$$d \mid 5.$$

Since  $d$  is a positive divisor of 5, the only possibilities are

$$d = 1 \quad \text{or} \quad d = 5.$$

It remains to verify that both values occur.

For  $n = 2$ ,

$$n^2 + 1 = 5, \quad (n + 1)^2 + 1 = 10.$$

Both 1 and 5 divide these numbers, so each candidate is attained.

Therefore the complete set of positive integers  $d$  is

$$\boxed{1, 5}.$$